

Lecture 04: Potential Outcomes

EEFE 530

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Schedule

Roadmap

Today's lecture:

- Define causal effects using the potential outcomes framework
- Introduce counterfactuals and the fundamental problem of causal inference
- Define ATE, ATT, and ATU
- Understand selection bias and bias decomposition
- Connect identification assumptions to common empirical strategies

Learning Objectives

By the end of this lecture, you should be able to:

- Define potential outcomes and observed outcomes
- State and interpret ATE, ATT, ATU
- Explain why causal effects are not directly observable
- Decompose selection bias in differences-in-means
- State the conditional independence and overlap assumptions
- Interpret regression and matching as identification strategies

Reading for next week

Potential outcomes

- Mixtape Ch. 4
- Effect Ch. 10
- *MHE* Ch. 2

Reading for next week

Matching

- Mixtape Ch. 5
- Effect Ch. 14
- *MHE* Ch. 3.2
- Matching sections in:
 - Imbens & Woolridge (2009)
 - Caliendo & Kopeinig (2008)
- Applications/Methods Software
 - Diamond and Sekhon (2013)
 - Sekhon (2011); Ferraro & Mirando (2017)
 - Imbens (2015)

Review

Research fundamentals

Angrist and Pischke (MHE) provide four fundamental questions for research:

1. What is the causal relationship of interest?
2. How would an ideal experiment capture this causal effect of interest?
3. What is your identification strategy?
4. What is your mode of inference?

Seemingly straightforward questions can be fundamentally unanswerable.

The experimental ideal

Why are experiments ideal?

Science widely regards randomized experiments as the gold standard for research.

But why? The costs can be substantial.

Costs

- slow and expensive
- heavily regulated by review boards
- can abstract away from the actual question/setting

Benefits

So the benefits need to be pretty large, right?

Example: Hospitals and health

Imagine we want to know the **causal effect of hospitals on health**.

Research question

Within the population of poor, elderly individuals, does visiting the emergency room for primary care improve health?

Empirical exercise

1. Collect data on *health status* and *hospital visits*.
2. Summarize health status by hospital-visit group.

Example: Hospitals and health (*MHE*)

MHE reports empirical exercise from the 2005 National Health Interview Survey:

Group	Sample Size	Mean Health Status	Std. Error
Hospital	7,774	3.21	0.014
No hospital	90,049	3.93	0.003

We get a *t* statistic of 58.9 when testing a difference in groups' means (0.72).

Conclusion? Hospitals make sick people sicker.

Alternative conclusion: Perhaps we're making a mistake in our analysis...

maybe sick people go to hospitals?

Any other other other explanations?

Potential outcomes framework

Let's develop a framework to better discuss the problem here.

- Binary treatment variable (e.g., hospitalized): $\mathbf{D}_i = \mathbf{0}, \mathbf{1}$
- Outcome for individual i (e.g., health): $\mathbf{Y}_i$

This framework has a few names...

- Neyman potential outcomes framework
- Rubin causal model
- Neyman-Rubin potential outcome/causal framework/model

Potential outcomes framework

Research question: Does $\mathbf{D}_i$ affect $\mathbf{Y}_i$?

For each individual i , there are two potential outcomes (w/ binary $\mathbf{D}_i$)

1. $\mathbf{Y}_{1i}$ if $\mathbf{D}_i = 1$

i 's health outcome if she went to the hospital

2. $\mathbf{Y}_{0i}$ if $\mathbf{D}_i = 0$

i 's health outcome if she did not go to the hospital

The difference between these two outcomes gives us the causal effect of hospital treatment, *i.e.*,

$$\tau_i = \mathbf{Y}_{1i} - \mathbf{Y}_{0i}$$

Q: What is τ_i ?

Fundamental problem

This simple equation: $\tau_i = Y_{1i} - Y_{0i}$

leads us to *the fundamental problem of causal inference*.

We can never simultaneously observe Y_{1i} and Y_{0i} .

The same individual cannot simultaneously **go** to the hospital and **not go** to the hospital. Most of applied econometrics focuses on addressing this simple problem.

Accordingly, our methods try to address the related question > For each Y_{1i} , what is a (reasonably) good counterfactual?

What determines “(reasonably) good” depends on our assumptions. This is where comparing to the experimental ideal is helpful.

Solutions?

Problem: We cannot directly calculate $\tau_i = Y_{1i} - Y_{0i}$.

Proposed solution: Compare outcomes for people who visited the hospital ($Y_{1i} \mid D_i = 1$) to outcomes for people who did not visit the hospital ($Y_{0j} \mid D_j = 0$).

$$E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0]$$

which gives us the *observed difference in average health outcomes*.

Q: This comparison will return *an* answer, but is it *the* answer we want?

Selection

Q: What does $E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0]$ actually tell us?

A: A helpful decomposition of Y_i is

$$Y_i = Y_{0i} + D_i \underbrace{(Y_{1i} - Y_{0i})}_{\tau_i}$$

Now write out our expectation, apply this definition, do creative math.

$$E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0]$$

$$= E[Y_{0i} + D_i (Y_{1i} - Y_{0i}) \mid D_i = 1] - E[Y_{0i} + D_i (Y_{1i} - Y_{0i}) \mid D_i = 0]$$

$$= E[\textcolor{blue}{Y}_{1i} \mid \textcolor{blue}{D}_i = \textcolor{blue}{1}] - E[\textcolor{green}{Y}_{0i} \mid \textcolor{blue}{D}_i = \textcolor{blue}{1}] + E[\textcolor{green}{Y}_{0i} \mid \textcolor{blue}{D}_i = \textcolor{blue}{1}] - E[\textcolor{green}{Y}_{0i} \mid \textcolor{green}{D}_i = \textcolor{green}{0}]$$

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Now write out our expectation, apply this definition, do creative math.

$$\begin{aligned} & E[Y_i \mid \mathbf{D}_i = 1] - E[Y_i \mid \mathbf{D}_i = 0] \\ &= E[\mathbf{Y}_{0i} + \mathbf{D}_i (\mathbf{Y}_{1i} - \mathbf{Y}_{0i}) \mid \mathbf{D}_i = 1] - E[\mathbf{Y}_{0i} + \mathbf{D}_i (\mathbf{Y}_{1i} - \mathbf{Y}_{0i}) \mid \mathbf{D}_i = 0] \\ &= \underbrace{E[\mathbf{Y}_{1i} \mid \mathbf{D}_i = 1] - E[\mathbf{Y}_{0i} \mid \mathbf{D}_i = 1]}_{\text{Average treatment effect on the treated} \text{ 😊}} + E[\mathbf{Y}_{0i} \mid \mathbf{D}_i = 1] - E[\mathbf{Y}_{0i} \mid \mathbf{D}_i = 0] \end{aligned}$$

Selection

Q: What does $E[Y_i \mid \mathbf{D}_i = 1] - E[Y_i \mid \mathbf{D}_i = 0]$ actually tell us?

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Selection

The **first term** is *good variation*—essentially the answer that we want.

$$E[Y_{1i} \mid D_i = 1] - E[Y_{0i} \mid D_i = 1]$$

$$= E[Y_{1i} - Y_{0i} \mid D_i = 1]$$

$$= E[\tau_i \mid D_i = 1]$$

The **average causal effect** of hospitalization *for hospitalized individuals* - Average Treatment Effect on the Treated (ATT).

The **second term** is *bad variation*—preventing us from knowing the answer.

$$E[Y_{0i} \mid D_i = 1] - E[Y_{0i} \mid D_i = 0]$$

Selection bias: The extent to which the “control group” provides a bad counterfactual for the treated individuals.

Q: Can you describe the selection bias ($E[Y_{0i} \mid D_i = 1] - E[Y_{0i} \mid D_i = 0]$) in our hospitalization example in words?

Selection

Angrist and Pischke (MHE, p. 15):

The goal of most empirical economic research is to overcome selection bias, and therefore to say something about the causal effect of a variable like $\mathbf{D}_i$.

Q: So how do experiments—the gold standard of empirical economic (and scientific) research—accomplish this goal and overcome selection bias?

Back to experiments

Q: How do experiments overcome selection bias?

A: Experiments break the link between potential outcomes and treatment. *In other words:* Randomly assigning $\mathbf{D}_i$ makes $\mathbf{D}_i$ independent of which outcome we observe (meaning $\mathbf{Y}_{1i}$ or $\mathbf{Y}_{0i}$).

Difference in means with random assignment of $\mathbf{D}_i$

$$E[\mathbf{Y}_i \mid \mathbf{D}_i = 1] - E[\mathbf{Y}_i \mid \mathbf{D}_i = 0]$$

$$= E[\mathbf{Y}_{1i} \mid \mathbf{D}_i = 1] - E[\mathbf{Y}_{0i} \mid \mathbf{D}_i = 0]$$

$$= E[\mathbf{Y}_{1i} \mid \mathbf{D}_i = 1] - E[\mathbf{Y}_{0i} \mid \mathbf{D}_i = 1] \text{ from random assignment of } \mathbf{D}_i$$

$$= E[\mathbf{Y}_{1i} - \mathbf{Y}_{0i} \mid \mathbf{D}_i = 1]$$

$$= E[\tau_i \mid \mathbf{D}_i = 1]$$

$$= E[\tau_i]$$

Random assignment of $\mathbf{D}_i$ breaks selection bias.

Randomly assigned treatment

The key to avoiding selection bias: **random assignment of treatment**

(or *as-good-as random assignment*, e.g., natural experiments).

Random assignment of treatment gives us

$$E[Y_{0i} \mid D_i = 0] = E[Y_{0i} \mid D_i = 1]$$

meaning the control group's mean now provides a good counterfactual for the treatment group's mean.

In other words, there is no selection bias, i.e.,

$$\text{Selection bias} = E[Y_{0i} \mid D_i = 1] - E[Y_{0i} \mid D_i = 0] = 0$$

Randomly assigned treatment

Additional benefit of randomization:

The **average treatment effect (ATE)** is now representative of the *population average*, rather than the *treatment-group average (ATT)*.

$$E[\tau_i \mid \mathbf{D}_i = 1] = E[\tau_i \mid \mathbf{D}_i = 0] = E[\tau_i]$$

Average Treatment Effect on the Untreated (ATU)

The **Average Treatment Effect on the Untreated (ATU)** answers:

*What would have happened to those who did **not** receive treatment if they **had** been treated?*

Formally:

$$\text{ATU} \equiv \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 0]$$

- $Y_i(1)$: potential outcome **with** treatment
- $Y_i(0)$: potential outcome **without** treatment
- $D_i = 0$: individuals who **did not** receive treatment

ATU is a **counterfactual effect for non-participants**.

Interpreting ATU

ATU is useful when asking **policy-relevant expansion questions**, such as:

- What would happen if we **expanded eligibility**?
- What are the gains from **inducing non-participants** to participate?
- How would outcomes change under **universal treatment**?

ATU generally differs from ATT when **selection into treatment is non-random**.

If treatment effects are heterogeneous:

$$ATE \neq ATT \neq ATU$$

Comparing ATE, ATT, and ATU

Estimand	Definition	Interpretation
ATE	$\mathbb{E}[Y_i(1) - Y_i(0)]$	Average effect in the full population
ATT	$\mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 1]$	Effect for those who received treatment
ATU	$\mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 0]$	Effect for those who did not receive treatment

These estimands coincide **only if treatment effects are homogeneous**.

With selection on gains:

$$\text{ATE} \neq \text{ATT} \neq \text{ATU}$$

Graphical Intuition: Sorting on Gains

Think of individuals as having **different treatment gains**:

$$\tau_i = Y_i(1) - Y_i(0)$$

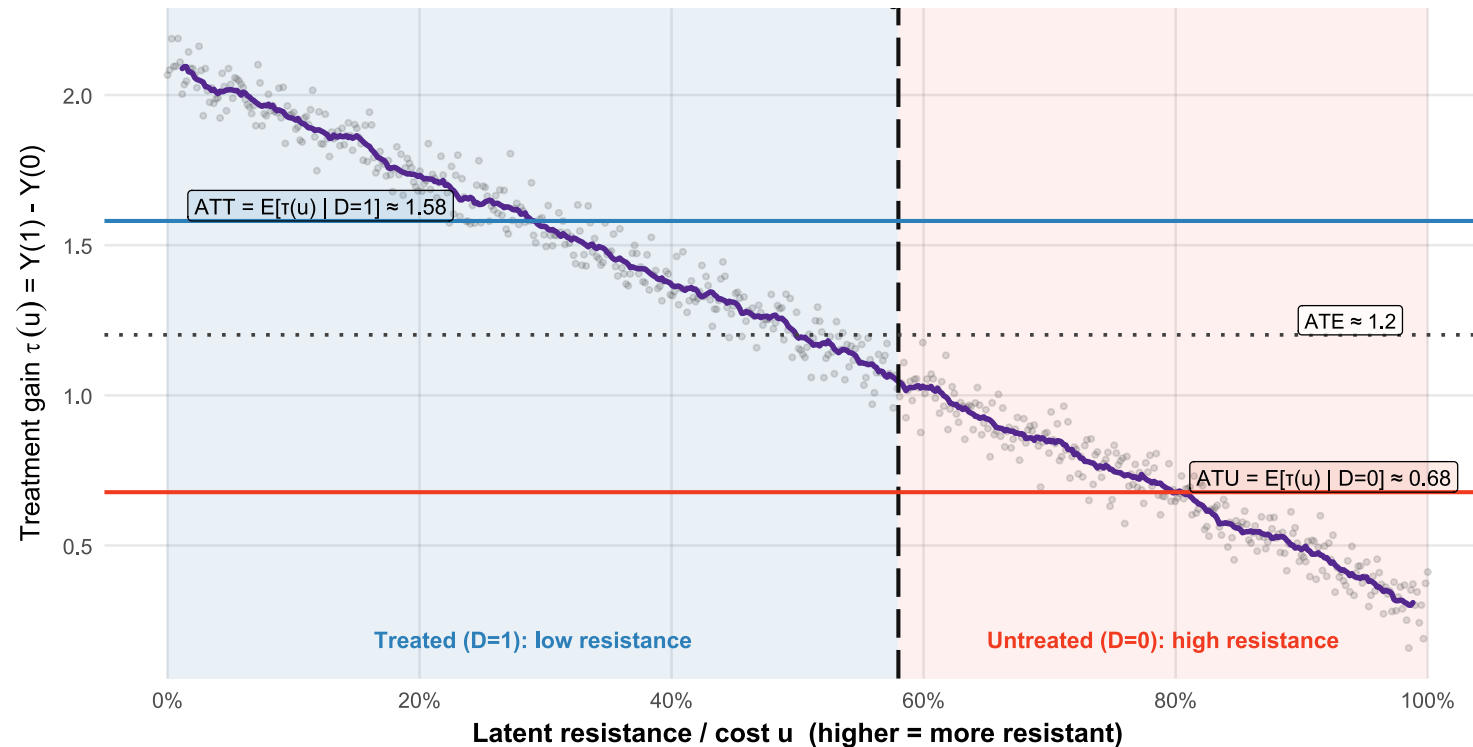
- Individuals with **high gains** are more likely to select into treatment
- Individuals with **low or negative gains** remain untreated

As a result: - **ATT** averages gains among *high- τ_i* individuals
- **ATU** averages gains among *low- τ_i* individuals

Selection bias arises because treatment status reveals information about potential outcomes.

Graphical intuition: Selection on gains via a latent trait

Selection on gains (sorting on a latent trait)



Selection on gains with heterogeneous treatment effects $\tau(u)$. Individuals with low latent resistance u select into treatment ($u \leq u^*$), so ATT averages higher gains than ATU.

Key point: if gains $\tau(u)$ vary with u and selection depends on u , then $ATT \neq ATU$ (sorting on gains).

Example: Energy efficiency programs

Governments subsidize energy efficiency investments (sometimes specifically targeting low-income homeowners)

Q: Do these programs have the desired effects (*i.e.*, lower energy use)?

A: Observational studies—comparing energy data from participants and non-participants—may not be a good approach].

Challenges: Participants self select. + Programs target lower-income homeowners.

Example: Energy Efficiency programs

How do we formalize these concerns in our framework?

Observational program evaluations:

$$\begin{aligned} E[\text{Energy}_i \mid \text{Program}_i = 1] - E[\text{Energy}_i \mid \text{Program}_i = 0] = \\ \underbrace{E[\text{Energy}_{1i} \mid \text{Program}_i = 1] - E[\text{Energy}_{0i} \mid \text{Program}_i = 1]}_{\text{Average causal effect of energy efficiency program on energy use for participants, i.e., } \bar{\tau}_1} + \\ \underbrace{E[\text{Energy}_{0i} \mid \text{Program}_i = 1] - E[\text{Energy}_{0i} \mid \text{Program}_i = 0]}_{\text{Selection bias}} \end{aligned}$$

If the program attracts/selects individuals who, on average, have higher energy use without the program (sort of the point of the program), then we have positive selection bias (decreasing estimated effect of the ATT in absolute value).

Q: Is this the only type of selection bias we should expect for this type of program?

Example: Energy Efficiency programs

$$\begin{aligned} E[\text{Energy}_i \mid \text{Program}_i = 1] - E[\text{Energy}_i \mid \text{Program}_i = 0] = \\ E[\text{Energy}_{1i} \mid \text{Program}_i = 1] - E[\text{Energy}_{0i} \mid \text{Program}_i = 1] + \\ E[\text{Energy}_{0i} \mid \text{Program}_i = 1] - E[\text{Energy}_{0i} \mid \text{Program}_i = 0] \end{aligned}$$

So even if the program, on average, has a negative impact on energy use (in the participant group), *i.e.*, $\bar{\tau}_1 > 0$, we will underestimate the magnitude if there is positive selection bias.

If the bias is sufficiently large (relative to the treatment effect), our estimate may even get the sign of the effect wrong.

More to come: We will go through an example in water conservation that highlights these challenges: [Brent & Ward \(2018\)](#)

The experimental ideal

Example: The STAR experiment (*MHE*)

The Tennessee STAR experiment is a famous/popular example of an experiment that allows us to answer an important social/policy question.

Research question: Does classroom size (or resources more generally) affect student performance?

- Statewide(-ish) in Tennessee for the 1985–1986 kindergarten cohort
- Ran for 4 years with ~11,600 children. Cost ~\$12 million.

Treatments:

1. *Small* classes (13–17 students)
2. *Regular* classes (22–35 students) plus part-time teacher's aide
3. *Regular* classes (22–35 students) plus full-time teacher's aide

First question: Did the randomization balance participants' characteristics across the treatment groups?

Ideally, we would have pre-experiment data on the outcome variable (test scores).

Unfortunately, we only have a few demographic attributes.

TABLE 2.2.1, MHE

Treatment: Class Size

Variable	Small	Regular	Regular + Aide	P-value
<i>Free lunch</i>	0.47	0.48	0.50	0.09
<i>White/Asian</i>	0.68	0.67	0.66	0.26
<i>Age in 1985</i>	5.44	5.43	5.42	0.32
<i>Attrition rate</i>	0.49	0.52	0.53	0.02
<i>K. class size</i>	15.10	22.40	22.80	0.00
<i>K. test percentile</i>	54.70	48.90	50.00	0.00

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Demographics appear balanced across the three treatment groups.

Maybe not for free lunch, but the differences are pretty small.

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The three groups differ significantly on attrition rate.

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The randomization generated variation in the treatment.

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The small-class treatment significantly increased test scores.

The experimental ideal

The STAR experiment

The previous table estimated/compared the treatment effects using simple differences in means.

We can make the same comparisons using regressions.

Specifically, we regress our outcome (test percentile) on dummy variables (binary indicator variables) for each treatment group.

Regression analysis

Assume for the moment that the treatment effect is constant[†], i.e.,

$$Y_{1i} - Y_{0i} = \rho \quad \forall i$$

then we can rewrite

$$Y_i = Y_{0i} + D_i (Y_{1i} - Y_{0i})$$

as

$$Y_i = \underbrace{\alpha}_{=E[Y_{0i}]} + D_i \underbrace{\rho}_{Y_{1i} - Y_{0i}} + \underbrace{\eta_i}_{Y_{0i} - E[Y_{0i}]}$$

[†]You'll often hear econometricians say "homogeneous" (vs. "heterogeneous").

Regression analysis

$$Y_i = \alpha + D_i \rho + \eta_i$$

Now write out the conditional expectation of Y_i for both levels of D_i

$$E[Y_i \mid D_i = 1] =$$

$$E[\alpha + \rho + \eta_i \mid D_i = 1]$$

$$= \alpha + \rho + E[\eta_i \mid D_i = 1]$$

$$E[Y_i \mid D_i = 0]$$

$$= E[\alpha + \eta_i \mid D_i = 0]$$

$$= \alpha + E[\eta_i \mid D_i = 0]$$

Take the difference... $E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0]$

$$= \rho + \underbrace{E[\eta_i \mid D_i = 1] - E[\eta_i \mid D_i = 0]}_{\text{Selection bias}}$$

Regression analysis

$$E[Y_i \mid \mathbf{D}_i = 1] - E[Y_i \mid \mathbf{D}_i = 0] = \rho + E[\eta_i \mid \mathbf{D}_i = 1] - E[\eta_i \mid \mathbf{D}_i = 0]$$

Again, our estimate of the **treatment effect** (ρ) is only going to be as good as our ability to eliminate the **selection bias**.

Selection bias in regression model: $E[\eta_i \mid \mathbf{D}_i = 1] - E[\eta_i \mid \mathbf{D}_i = 0]$

Selection bias here should remind you a lot of

omitted-variable bias. There is something in our disturbance η_i that is affecting $\mathbf{Y}_i$ and is also correlated with $\mathbf{D}_i$.

In other econometrics parlance: Our treatment $\mathbf{D}_i$ is *endogenous* if it is correlated with the error term η_i .

Solutions and covariates

Selection bias in regression model: $E[\eta_i | \mathbf{D}_i = 1] - E[\eta_i | \mathbf{D}_i = 0]$

As before, if we randomly assign $\mathbf{D}_i$, then selection bias disappears.

Another potential route to identification is to condition on covariates in the hopes that they “take care of” the relationship between $\mathbf{D}_i$ and whatever is in our disturbance η_i .

Without very clear reasons explaining how you’ve controlled for **all** the “bad variation”, clean and convincing identification on this path is going to be challenging.

Covariates

That said, covariates can help with two things:

1. Even experiments may need **conditioning/controls**: The STAR experiment was random *within school*—not across schools.
2. Covariates can reduce residual variance—increasing precision.

Now that we've seen regression can analyze experiments, let's estimate the STAR example...

TABLE 2.2.2, MHE

Explanatory variable	1	2	3
<i>Small class</i>	4.82 (2.19)	5.37 (1.26)	5.36 (1.21)
<i>Regular + aide</i>	0.12 (2.23)	0.29 (1.13)	0.53 (1.09)
<i>White/Asian</i>			8.35 (1.35)
<i>Female</i>			4.48 (0.63)
<i>Free lunch</i>			-13.15 (0.77)
<i>School F.E.</i>	F	T	T

The omitted level is *Regular* (with part-time aide).

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<i>Free lunch</i>			-13.15 (0.77)
<i>School F.E.</i>	F	T	T

Results without other controls are very similar to the difference in means.

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<i>School F.E.</i>	F	T	T

School FEs enforce the experiment's design and increase precision.

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<i>School F.E.</i>	F	T	T

Additional controls slightly increase precision.

Future work

Future work

The rest of the course will focus on methods and designs to eliminate selection bias and estimate causal effects.

We will try to think about these in the context of the causal framework.

Some of these that you should be comfortable implementing (by the end of the course) are:

- Instrumental variables
- Matching
- Difference-in-difference designs
- Synthetic control
- Regression discontinuity